

Temporal Exploit Prediction — Progress Update

Predicting *when* a published CVE becomes weaponized (survival analysis; complements EPSS) · week of 23–30 June 2026
· supervisor meeting, 1 July 2026

Headline. The in-wild head was re-targeted into a *measurable* comparison against EPSS and adversarially verified: at disclosure (cold-start), structured publication-time features out-rank EPSS by **+0.100 AUC@30** (95% CI [0.055, 0.145], wins 13/15 time-splits); PR-AUC is tied. The whole project is now consolidated into a 19-page, fully-cited report.

What's new since last week

- **A concrete win over EPSS at $t=0$.** On 1,310 in-wild test events across 15 walk-forward origins: structural AUC@30 **0.795** vs raw-EPSS **0.695** ($\Delta+0.100$); AUC@90 $\Delta+0.134$ (wins 15/15); top-decile recall **0.47** vs 0.27. Value = the cold-start regime where EPSS is near-chance / ~48% missing.
- **GreyNoise telemetry connector built** (verified API, TDD, 6 new tests, passed an RE audit) — the next lever to grow scarce in-wild labels. Limitation surfaced: it is *prospective-only* (rolling $\leq 30d$, no historical backfill).
- **Report is now literature-grounded:** 40 verified citations + References section integrated into the 19-page PDF; sources checked against DOI / arXiv / venue.

Why the result is trustworthy (rigor corrections)

- **Fixed a baseline bug that would have inflated the win.** EPSS must be ranked by its *raw* score: wrapping it in a model collapses it to chance (AUC@30 **0.501** vs raw 0.695). The +0.100 win is measured against a *strong* EPSS.
- **Corrected the target framing (honesty).** The “in-wild” label is ~93% VulnCheck catalog-add dates — an administrative proxy (median ~+175d lag; ~22% within 30d), not true in-the-wild onset. Claim is “cold-start ranking advantage,” not “predicts real-world exploitation.” Adversarial RE (2 agents) found no leakage or handicap.
- **Caught 3 errors in the report's own draft citations** while adding references (a wrong year, a samples-vs-events unit error, a wrong journal volume); flagged 1 non-peer-reviewed source.

Decisions to discuss

- **Framing:** is “cold-start ranking advantage over EPSS” the right thesis claim, given the proxy label (+175d lag)?
- **Telemetry:** GreyNoise is forward-only — wire it for prospective evaluation, or accept the retrospective ~396-event data ceiling as the wall?
- **Dissertation angle:** continue toward the epidemiological mixture model for pre-disclosure dynamics?
- **Logistics:** merge the inwild-epss-parity branch to master.

Status: 337 tests green · report 19pp / 40 refs · parity work pushed (branch inwild-epss-parity)	Sources: artifacts/inwild_epss_parity.json; commits 79d4213, cf3348d, 872b588 (parity), bace8e8/fdb2541 (GreyNoise); docs/project_report_2026-06-23.pdf
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